

1. Check the convergence of the series

$$\sum_{k=1}^{\infty} \left(\sqrt{k+1} - \sqrt{k} \right)^{-1/2} \ln \frac{k-1}{k+1}.$$

2. Check the uniform convergence of the functional sequence

$$f_n(x) = e^{n(x-1)}$$

on $E = (0, 1)$.

3. Check the uniform convergence of the series

$$\sum_{k=1}^{\infty} \frac{x}{1 + n^4 x^3}$$

on $E = (0, +\infty)$.

4. Find Maclaurin series of the function

$$f(x) = \frac{1}{4} \ln \frac{1+x}{1-x} + \frac{1}{2} \arctan x.$$

5. Find the radius of convergence and the sum of the power series

$$\sum_{k=1}^{\infty} k(k+2)x^k.$$